Bocconi

TECHNOLOGY RISK GOVERNANCE. MAJOR ASSIGNMENT

Instructions

Università Bocconi MILANO

POLITECNICO MILANO 1863

Course written assignments

- Students are asked for preparing a **mandatory (group) major assignment** selecting one of the following options:
  - **Critical Incident Analysis**: description and in-depth analysis of an industrial accident/disaster involving cyber or IT risk factors
  - **Technology Risk Assessment**: perform a preliminary risk assessment of a digital solution for a specific product/system/process of your interest
- To improve your final mark, you may **voluntary** complete one or two **(group) minor assignments**:
  - Analysis of the accident in a Co-generation Plant (Class 9)
  - Critical discussion of a critical incident in an interdependent networked infrastructure system (Class 19).
- Submission deadlines:
  - Minor Assignment reports: by Monday, Dec 22, 2025
  - Major Assignment report: at least one week before the oral exam session you want to attend

Objectives and Deliverable

- The assignment is intended to deepen your capability of critically apply Technology Risk Governance principles and methods
- Deliverable: prepare a **report of about 15 pages**
- Groups of **max three (3) students** are accepted



Critical Incident Report

AIM AND STRUCTURE

Students should identify a relevant industrial accident/disaster involving cyber or IT risk factors, for which enough secondary data are available from different sources (technical and scientific literature)

All the industrial and service sectors are eligible.

— The paper should cover this core contents:

- **Description** of the company/system of interest and of the event, based on publicly available documentation
- **Model-based in-depth analysis** of the accident by applying the most appropriate technique, considering both technical and organizational factors
- **Critical discussion** of the appropriateness of the implemented countermeasures, and generalization of the lesson learned to improve safety performance in the specific industry

Technology Risk Assessment report

AIM AND STRUCTURE

Referring to a specific industrial (e.g. manufacturing, process industry, transportation, etc.) or service (e.g. banking, healthcare, transport) domains students should select an existing or emerging digital solution at either product, system o process level, and perform a preliminary risk assessment using secondary data

— The paper should cover this core contents:

- **Introduction**. Brief description of the area of investigation (industry/ies and business model, product, system, processes)
- **Business case**. Description of the technological solution and the main specifications for implementation (or main use cases)
- **Reference documentation**. Search methodology and the knowledge base selected for the analysis
- **Risk Assessment**. Selection and justification of the RA method. Reporting of all the Risk Assessment phases (identification/analysis/ evaluation)
- **Discussion**. Suggest possible risk mitigation options and assess residual risk

About the use of GenAI

The final report should transparently report about the use of GenAI.

This information should be reported and justified:

- In the Introduction or Methodology Sections
- With a specific GenAI Disclosure Statement appended to the report
  (see the next slide)

General principles

- Gen AI is a powerful tool in the hands of engineers and managers
- Under no circumstances should we allow GenAI to guide our actions and make decisions on our behalf

Permitted and encouraged uses of GenAI:

- Support the collection and critical analysis of relevant information and data
- Encoding of heterogeneous information/data within a framework provided by the analyst
- Development of an AI agent that implements the solution developed by the analyst
- Proof reading and language sanity check of the final report

Prohibited uses of GenAI:

- Selection of study contents and methodology
- Selection of existing frameworks or methods
- Development of new analytical frameworks

GenAI Disclosure Statement

(to be appended to the final report and signed by all the team members)

The project and the final report were developed with the assistance of generative AI (GenAI) tools. The following details the extent and nature of AI involvement:

- **GenAI Tool(s) Used:** [Specify the tool(s), e.g., 'ChatGPT-4o' or 'Google Gemini'].
- **Purpose of Use:** [Describe how AI was used, e.g., 'for brainstorming initial ideas', 'generating outlines', 'drafting sections', or 'editing for clarity'].
- **Specific Contributions:** [Detail AI-generated content, e.g., 'AI assisted in summarizing literature on topic X, which informed the benchmarking of Risk Assessment methos and the final selection'].
- **Human Oversight:** All AI outputs were reviewed, edited, and verified by the authors for accuracy, originality, and relevance. No AI-generated content was submitted verbatim without substantial modification.
- **Ethical Considerations:** This usage aligns with Politecnico di Milano's policy and guidelines. The authors take full responsibility for the final content and attest that it represents their own intellectual effort.

NOTE: If relevant, some prompts could be included in Annex for methodological transparency and replicability

Deadlines for the Major Assignment

- On line review sessions on **Thursday 10/12/2026** and **Tuesday 15/12/2026**.
- **Composition** of the group (names and contacts) the type and a **tentative title** of the report must be communicated by email to paolo.trucco@polimi.it no later than **Friday 13/11/26**
- The **final manuscript** must be submitted at least **one week before** the oral exam call you want to attend
- **Submission**: submit a digital copy (Word or PDF) of your final report on WeBeep

Useful Resources

- Google Scholar, https://scholar.google.com/
- Scopus, https://www.scopus.com/search/form.uri?display=basic
- ProQuest, https://www.proquest.com/index

- POLIMI and Bocconi Libraries and electronic resources

- Books and other documents that you may ask for to the instructors

THANKS.



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